

JUAN PABLO ALVAREZ OVALLE

Backend Developer · Node.js / TypeScript · IoT & Systems Integration

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SUMMARY

Backend developer with 6+ years of experience building hardware-software integration and instrumentation solutions using a modern stack (Node.js, TypeScript, Express, PostgreSQL, MongoDB, Redis), with a focus on REST APIs, systems integration, and real-time communication (MQTT, WebSockets).

Electronic engineer specialized in instrumentation, with experience in automation and IoT for clients in the industrial and laboratory sectors, and a strong foundation in application architecture, system interoperability, and complex technical problem-solving. Trilingual: Spanish (native), English (B2), French (B2). Currently building expertise in automated testing and modern DevOps practices with Docker.

SKILLS

Backend and real-time: Node.js, Express, TypeScript, JavaScript, REST APIs, WebSockets (Socket.io), Pub/Sub, Event-driven

Databases: PostgreSQL, MongoDB, SQLite, Redis, ORMs: TypeORM, Mongoose

Industrial integration and IoT: MQTT, Modbus TCP, OPC, PLC communication, National Instruments hardware, LabVIEW, sensor data acquisition

Containerization: Docker (Dockerfiles, images, Docker Compose)

Frontend (supporting skills): React, Electron

Version control: Git, GitHub

Expanding expertise in: Automated Testing (Jest)

PROFESSIONAL EXPERIENCE

Software Consultant – Backend Developer – Freelance *Feb 2018 – Present*

Directly for Veolia Aguas de Tunja and Ingemet I&C S.A.S. As a subcontractor for Kuspyde Ingeniería and its final customers (Empresa de acueducto y alcantarillado de Bogotá E.S.P., EPM, Hidrométrica S.A.)

- Developed a suite of 3 calibration and testing systems for water meters, compliant with applicable standards, using Node.js, TypeScript, React, Electron, PostgreSQL and LabVIEW, enabling new laboratory capabilities such as homologation testing and large-diameter calibration.
- Owned the full lifecycle of each system — requirements, development, testing, on-site installation, and delivery — and automated the workflow from order intake to certificate generation, eliminating manual errors and reducing operational time.
- Developed the backend of an IoT dashboard that gathers real-time sensor data via MQTT and REST APIs. The results are distributed using Redis pub/sub and Socket.io, and presented in a React web app that is currently under development.
- Integrated PLCs, industrial instrumentation, and National Instruments hardware with software systems — via Modbus TCP/RTU, OPC, and NI-DAQmx/VISA drivers in LabVIEW — for real-time data acquisition, control, and process monitoring.
- Maintained the operational stability of production applications by resolving technical issues within a sub-24-hour SLA, earning client trust for long-term, continuous support.

Product Engineer *Jun 2014 – Feb 2018*

Kuspyde Ingeniería S.A.S.

- In my final year, I took technical leadership of the software development area, autonomously delivering 3 hardware-software integration projects on schedule.
- Built complete software applications in LabVIEW for industrial measurement and process control, including user management, data persistence, and report generation.
- Implemented PID control for a thermal stress-relief system used on hydroelectric turbine parts, regulating multi-hour heating and cooling ramps via thermocouples and MOSFETs, maintaining temperature within a 3 °C margin.

- Reduced production costs by replacing a failure-prone handheld device with an in-house Android app (TCP communication), deployed across 4 projects.
- Automated labor-metrics evaluation using Google Sheets and Apps Script, eliminating 30+ hours of manual work per month.

EDUCATION

Postgraduate Specialization in Electronic Instrumentation

2012–2013

Santo Tomás University

Electronic Engineering

2006–2012

Santo Tomás University

LANGUAGES

Spanish (native speaker), English (B2), French (B2)

PROFESSIONAL DEVELOPMENT

- Node.js, Express, MVC, REST APIs, GraphQL — Udemy
- JavaScript and TypeScript — Udemy
- React, Redux, React Router — Udemy
- Web and Backend Development — Platzi
- Docker, Kubernetes — in progress (Udemy)